XFX GeForce 6600GT 128 MB

The culprit in the case of the ATi cards is the crippled clock speed and pixel pipelines. Another factor is the older architecture that the ATi cards are based on, as compared to the 6XXX series from nVidia, which has been developed from the ground up. The architectural enhancements do play a major role in the scores that the 6600 cards posted.

The 6600GT definitely costs more, but you can be sure of the performance of the card, which is much better in terms of frame rates and the eye candy possible in games—even at high resolutions. In fact, it sometimes beat the plain vanilla X800 in the game tests.

There is a continuous stream of new games released each year, and with each game the hardware ante is upped, at least in the video card department. Any investment on a 6600GT is money well spent—it will definitely stand you in good stead for at least the next two years.

Conclusion

The XFX GeForce 6600GT duo was the king here. No other card could come close to the scores posted by these two, in any game, be it OpenGL- or DirectX-based. They even topped the 3DMark scores. The surprising thing was that the XFX GeForce 6600GT 128 MB card posted scores identical to its 256 MB counterpart in all the games! It seems none of the games or benchmarks could utilise the extra 128 MB available on the XFX GeForce 6600GT 256 MB card.

In sum, the XFX GeForce 6600GT 256 MB wins the *Digit* Best Buy Gold award in the mid-range PCI-e card category, while the XFX GeForce 6600GT 128 MB wins the Digit Best Buy Silver award.

HIGH-END CARDS

This is where we talk about the cream of the crop! Low-end cards are accessible to anyone, and anyone willing to spend a little extra can get mid-range cards. However, this is the section where we talk about the Porsches and the Maybachs of the desktop graphics world.

Features

Features-wise, all these cards give you everything, including games, VIVO connectors and all the other stuff you could possibly expect. The VIVO connectors are especially mentionable, since you can directly provide component output to your HDTV from the card. Imagine playing *Doom 3* on your HDTV!

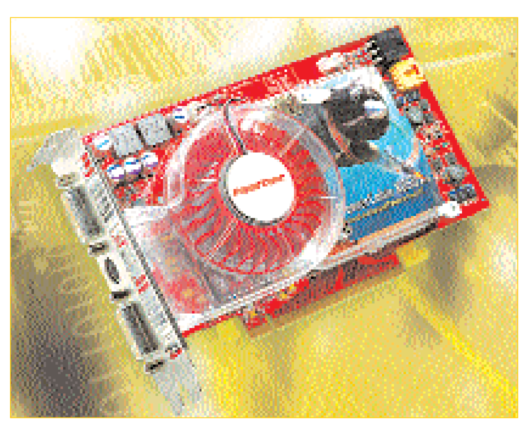
PowerColor and Asus were the two manufacturers that provided HDTV VIVO connectors. Games were aplenty, and MSI actually provided five games with the RX800. It's a pity the card couldn't make it to the comparison. On the other hand, XFX with their eye-catching X-shaped box provided the regular wires and cables—there was no HDTV connector. In terms of games, they bundled the full version of *FarCry* (DVD) with their high-end cards, which is sure to attract users.

Performance

We received 10 cards in this category, discounting the Asus 6800 Ultra Dual GPU monster and the SLI-setup. Of these, we could only review nine, since the MSI RX800 provided a corrupted display, and we could do nothing about it. The remaining nine are the best to date, anywhere, worldwide. These included top-of-the-line offerings from ATi and nVidia.

While nVidia has come out with two revisions of the graphics cards in their arsenal, ATi has stayed put with their X800 series. Sadly enough, the current nVidia cards make the X800 look and feel aged.

All the cards except the Gigabyte had active cooling circuits. Gigabyte chose to walk a different path and provided a passive pipe cooling



PowerColor X850 XT Platinum Edition 256MB

nVidia TurboCache And ATi HyperMemory

The nVidia GeForce 6200TC and the ATi X300 chips in the low-end segment featured new memory management techniques. While nVidia featured TurboCache, ATi featured HyperMemory. The bottomline is that both these implementations are in many respects similar to each other. In both the cases, the graphics card has dedicated memory onboard, but in addition to that, it can access a limited amount of system memory. This is something like AGP on a PCI-Express bus, because AGP cards can access system memory in a similar way, and this was known as AGP texturing. Using this method, the requirement of the amount of onboard memory is greatly reduced, and this contributes to reduction of price.

The concept is somewhat similar to motherboards with onboard graphics, but the difference is that with graphics cards, there is at least some amount of dedicated memory, whereas in the on-board graphics of motherboards, there is usually none. The advantage of TurboCache and HyperMemory is that they bring most of the features of high-end cards within the reach of the general populace. This technology provides the highest graphics performance-to-price ratio.